

The Binary Opposition and Dialectical Unity between the Disciplinary Essence and the Connotation of PBL: Theoretical Clarification and Practical Orientation of English PBL in Primary School

Ruoyu Wang¹, Li Jiang², Shuyan Wang³

1.College of Education, Zhejiang Normal University/Diehu Primary School, Zhengzhou Jingkai District, China, wangruoyubb@qq.com

2.College of Education, Zhejiang Normal University/ Zhengzhou No.36 Senior High School, China, jlzjnu36@yeah.net;

3.College of Education, Zhejiang Normal University/New Campus of Tai'an No.3 Senior High School, China, zjnusy07@163.com

Abstract:Project-based Learning (PBL) in primary school English often oscillates between two extremes: reducing PBL to contextualized language drills or designing inquiry tasks disconnected from learners' proficiency levels. This tension stems from a fragmented conceptualization of the relationship between the essence of English as a discipline and the intended outcomes of PBL. Grounded in Herbart's subject-centered theory, Second Language Acquisition (SLA) research, and dialectical materialism, this study clarifies the interdependence and dynamic equilibrium between language and inquiry. A four-pronged practical framework is proposed, encompassing cognitive restructuring, design anchored in disciplinary essence, process-oriented assessment, and differentiated scaffolding. A concrete instructional case illustrates the framework's applicability. The study aims to restore the pedagogical integrity of English PBL, fostering the synergistic development of linguistic competence and holistic literacy.

Keywords:Primary School English; Project-Based Learning (PBL); Disciplinary Essence; Binary Opposition; Implementation Pathway

1. Introduction

The Compulsory Education Curriculum Program (2022 Edition) explicitly advocates

* Corresponding Author: Li Jiang (jlzjnu36@yeah.net;)

“exploring integrated teaching methods such as thematic and project-based learning,” positioning Project-based Learning (PBL) as a core vehicle for curriculum reform and the cultivation of core competencies [1]. In English language education, PBL holds the promise of bridging the long-standing gap between knowledge learning and real-world application by immersing students in authentic contexts and sustained inquiry, an orientation aligned with the Compulsory Education English Curriculum Standards (2022 Edition)’s call for “integrating learning with thinking and application with creation” [2]. However, classroom practice at the primary level reveals a persistent “language-inquiry” binary opposition. On one hand, some teachers treat English merely as a set of linguistic forms, degrading PBL into contextualized drill activities devoid of genuine inquiry. On the other hand, some teachers overextend the inquiry component by directly borrowing empirical investigation models from science subjects, assigning tasks that far exceed students’ language capacity, so that the inquiry proceeds in the mother tongue while English is relegated to a decorative role in final products.

Existing research has probed PBL implementation in various disciplines [3][4], yet few studies have examined the philosophical tension between disciplinary essence and pedagogical approach specifically in the context of primary school English [5]. Recent reviews (e.g., 2022–2025) highlight the need for more discipline-specific PBL frameworks, but a systematic investigation of the dialectical relationship between language and inquiry remains absent. To fill this gap, this paper first problematizes the binary opposition as a cognitive and practical fallacy, then offers a theoretical synthesis informed by Herbart’s discipline-centered theory, second language acquisition (SLA) theories, and dialectical materialism, and finally proposes a practical pathway framework aimed at restoring the integrity of English PBL. The study is expected to benefit curriculum designers and frontline teachers seeking to implement PBL that authentically develops both language and inquiry skills.

2. Instructional Alienation: Practical Manifestations of the Binary Opposition in English PBL of Primary School

The binary opposition in English PBL of primary school is not merely an operational difficulty but a reflection of a deeper cognitive dislocation. Two distinct yet homologous types of alienation can be identified.

The first is a language-supremacy training orientation. Here, the essence of English is narrowed to the transmission of discrete linguistic knowledge, and PBL is reduced to a vehicle for situational drills. Driving questions become “How can I use the target sentence pattern to complete this dialogue?” Collaborative inquiry turns into role-based recitation, and final products are fill-in-the-blank worksheets. The defining features of PBL—complex authentic problems, generative inquiry, and higher-order thinking—are obscured; only mechanical language repetition remains. Both the communicative essence of English and the inquiry kernel of PBL are thus dissolved.

The second is an inquiry-supremacy de-disciplinization orientation. In this scenario, teachers import the empirical inquiry logic from science or engineering without adapting it to the linguistic demands of the English classroom. Tasks such as “using English to investigate an ecosystem” or “verifying a scientific principle in English” are designed, but students lack the language resources to undertake them meaningfully. Consequently, inquiry is conducted almost entirely in the mother tongue, and English is inserted only as a superficial overlay during presentations. English PBL degenerates into inquiry without language, losing the disciplinary core of language application and falling into formalism.

These two orientations, although seemingly opposite, share a common root: a fragmented understanding of the relationship between the disciplinary essence of English and the connotation of PBL. The former treats PBL as a tool for language training; the latter treats English as mere decoration for inquiry. Both violate the dialectical logic that “disciplinary essence provides the substance, and PBL connotation provides the function.”

3. Theoretical Foundation: The Value Legitimacy of the Nature of the Discipline and the Connotation of PBL

3.1. Disciplinary Essence: The “Ontological Basis” for Transforming Teaching Methods

Disciplinary essence constitutes the essential boundary and direction for any pedagogical reform [6]. Herbart’s discipline-centered theory posits that instruction must proceed from the “internal logic” of the discipline, and any methodological innovation must serve the discipline’s unique educational value [7]. For English, this essential stipulation is a tripartite unity of “linguistic instrumentality, cognitive mediation, and cultural carrier” [8][9]. Its core educational value is not the accumulation of vocabulary and grammar, but the ability to use language to convey meaning, develop thinking, and understand diverse cultures in authentic contexts. This nature mandates that reforms—whether optimizing traditional instruction or adopting PBL—must be gauged by whether they promote the transformation from “symbolic memory” to “communicative competence.”

Contrastingly, natural science is an empirical knowledge system suited to hypothesis-verification logic, and Chinese language arts is a humanistic textual system suited to interpretation-expression logic. Copying these inquiry models directly into English PBL without considering the discipline’s linguistic essence inevitably causes “disciplinary misplacement,” the theoretical root of the de-disciplinarization dilemma.

3.2. The Connotation of PBL: The “Methodological Support” for English Learning

PBL, as a paradigm “driven by authentic problems and aimed at competency development,” aligns intrinsically with English learning. From an SLA perspective, Krashen’s Comprehensible Input Hypothesis emphasizes meaning negotiation in authentic contexts [10], and Swain’s Output Hypothesis underscores the necessity of pushed language production for internalization [11]. PBL’s authentic driving questions create “situational anchors” for comprehensible input at the $i+1$ level, while collaborative inquiry generates genuine communicative needs for language output, facilitating the transformation of declarative knowledge into procedural ability.

From the perspective of educational objectives, Bloom's revised taxonomy identifies "analysis, evaluation, and creation" as higher-order cognitive goals [12], and the Core Competencies for Chinese Student Development highlights practical innovation and communication [13]. PBL's sustained inquiry and diverse product output engage students in these higher-order processes while fostering collaboration and communication—objectives that directly support the English curriculum's goals of developing language competence, thinking quality, and learning ability in tandem.

4.Philosophical Perspective: The Essence of the Binary Opposition and the Logic of Dialectical Unity

The "language-inquiry" binary opposition is, in essence, an imbalance between instrumental and value rationality. Dialectical materialism's law of the unity of opposites and Heidegger's ontology provide a penetrating lens for understanding and resolving this opposition.

4.1.The Philosophical Essence of the Binary Opposition: Metaphysical Thinking Dichotomy

The opposing tendencies of "language supremacy" and "inquiry supremacy" are concrete manifestations of metaphysical "either/or" thinking. They mistakenly treat language and inquiry as mutually exclusive entities, ignoring the universal connection and internal contradiction they form in PBL.

From a dialectical materialist perspective, language and inquiry constitute a unity of opposites. Language is the instrumental dimension (how to express in English), while inquiry is the procedural dimension (how to solve authentic problems). However, their unity is more fundamental: language serves as the material carrier for inquiry, and inquiry provides the meaningful field for language application. Their opposition is a difference within the same contradiction, not an absolute dichotomy. Metaphysical thinking severs this internal relation, leading to the two alienated instructional forms described earlier.

From an ontological perspective [14], English learning is a mode of human existence—connecting with the world through language. PBL is the practical form of this existence. "Language-supremacy" practice reduces learners to operators of linguistic symbols,

neglecting the existential need to understand the world through inquiry. “Inquiry-supremacy” practice reduces learners to problem-solving tools, neglecting the disciplinary attribute of confirming existence through language. Both deviate from the unity of “human-language-world” and cause the “existential meaning” of English learning to be lost.

4.2.The Theoretical Logic of Dialectical Unity: Reconstruction through Relational Thinking

Resolving the opposition requires reconstructing cognition through relational thinking, establishing a unified logic of “disciplinary essence as substance, PBL connotation as function.” This unity operates through three dimensions:

- **Interdependence:** Language and inquiry presuppose each other. Language devoid of inquiry is symbol manipulation; inquiry devoid of language in the English classroom loses disciplinary grounding. As Marx noted, essence exists in the ensemble of relations [15].
- **Mutual Transformation:** Inquiry generates language needs, prompting the proceduralization of declarative knowledge; enhanced language ability deepens inquiry, enabling higher-order analysis. A virtuous cycle of “inquiry promoting language, language strengthening inquiry” is formed.
- **Dynamic Equilibrium:** Emphasis should be adjusted according to learners’ grade level and proficiency. In lower grades, language scaffolding supports inquiry; in higher grades, inquiry demands drive language development. This adjustable, adaptive balance avoids the extremes of rigidity and vacuity.

5. Practical Orientation: Implementation Pathways for Dialectical Unity

5.1.Cognitive Reconstruction: Establishing a “Dual-Dimension Unified” Teaching Philosophy

This is the prerequisite pathway. Teachers must undergo a tripartite cognitive shift: from

opposition to symbiosis of language and inquiry (tool and field); from copying other disciplines' PBL models to adapting based on the English discipline's essence; and from single-objective orientation to synergistic competency development. Specialized training and collaborative case analysis workshops are recommended to effect this shift from metaphysical to dialectical thinking [16].

5.2.Design Anchoring: Constructing a PBL Plan Adapted to "Essence- Connotation"

Design must anchor both the disciplinary essence and the connotation of PBL. All inquiry activities must revolve around language application, and all language activities must carry inquiry attributes. Language knowledge is embedded dynamically according to inquiry needs, not pre-taught in isolation. The binding of "problem-language-thinking" ensures that inquiry is supported by language and that language use possesses inquiry significance [17].

Illustrative Case: "Our School Welcome Guide for International Visitors."

Core Driving question: "How can we create a useful English guide to help international visitors navigate our school and feel welcome?"

Inquiry process:

1. Students investigate key locations and interview staff (using scaffolded English question templates);
2. collaboratively design a map and write descriptive English texts;
3. peer-review drafts using a language checklist;
4. produce a bilingual booklet or digital presentation.

Language objectives: giving directions (turn left, go straight), describing locations (next to, in front of), making polite suggestions (you might want to...).

Product: a printed guidebook and a video tour. Language and inquiry are inseparable throughout.

Case-based dialectical analysis: This design operationally transcends the binary opposition

identified earlier. On the one hand, it avoids the “language-supremacy drill” trap—language objectives (locative expressions, polite suggestions) are not taught in isolated pre-learning sessions but are dynamically embedded within the authentic task of creating a practical guide. Students use English to solve the genuine problem of “how to make visitors feel welcome,” thereby endowing language with communicative vitality rather than reducing it to mechanical repetition. On the other hand, it circumvents the “inquiry-supremacy de-disciplinization” risk—the entire investigation, mapping, drafting, and peer-review process is mediated primarily through English, with teachers providing i+1-level comprehensible input scaffolds (e.g., question templates, language checklists) to ensure that inquiry unfolds alongside linguistic construction, rather than relegating English to a decorative overlay on final products. More fundamentally, this case embodies the threefold logic of dialectical unity: the investigative needs generate authentic communicative demands for language use (interdependence); the process of drafting descriptive texts and conducting peer reviews pushes students to refine their expression, which in turn deepens the logical organization of their spatial reasoning (mutual transformation); and the teacher’s dynamic adjustment of scaffold density—heavier linguistic support during initial interviews, gradually withdrawn during final production—maintains a context-sensitive dynamic equilibrium between language and inquiry across different phases. Consequently, the case actualizes the synergistic effect wherein “the driving problem initiates inquiry, inquiry activates language use, and language use deepens cognitive thinking,” ensuring that language and inquiry remain inseparable throughout.

5.3.Evaluation Innovation: Establishing a “Process-Competency” Oriented Evaluation System

Breaking the “product-only” limitation, a three-dimensional evaluation index is built:

- 1.Language Application: authentic, situational use of English, not mere accuracy;
- 2.Inquiry Process: depth of cognitive engagement, not just task completion;
- 3.Competency Synergy: integration of language, thinking, and culture.

To enhance operability, a sample rubric is provided in Table 1.

Table 1. Sample PBL Assessment Rubric for Primary School English

Dimension	Indicator	4 (Excellent)	3 (Proficient)	2 (Developing)	1 (Beginning)
Language Application	Appropriacy & fluency in oral/written tasks	Confidently uses target language with flexibility in new contexts	Uses target language with minor errors; meaning clear	Relies on formulaic expressions; communication sometimes unclear	Fragmented language; frequent breakdown in communication
Inquiry Process	Quality of questions, reasoning, and use of evidence	Asks probing questions; supports ideas with logical reasons in English	Asks relevant questions; provides simple reasons	Questions are superficial; reasons are weak or absent	Does not engage in inquiry; merely repeats given information
Competency Synergy	Demonstration of intercultural understanding & collaboration	Actively integrates cultural knowledge; facilitates group interaction	Shows awareness of cultural differences; cooperates well	Limited cultural reference; passive in collaboration	No cultural awareness; works in isolation

In addition, process portfolios and self/peer assessment checklists supplement the rubric to capture the learning journey.

5.4. Support Assurance: Building Collaborative “Language-Inquiry” Scaffolding

Scaffolding must be grade-differentiated to achieve dynamic balance. For lower grades (Grades 1–2), heavy reliance on language scaffolds such as picture-word cards, sentence starters, and model dialogues is needed. For middle grades (Grades 3–4), thinking-language integration scaffolds (e.g., graphic organizers with logical connectors) help students organize thoughts. For higher grades (Grades 5–6), inquiry scaffolds (e.g., question formulation techniques, data collection sheets) are foregrounded, while language scaffolds are gradually withdrawn. Scaffolding is a temporary bridge, not a permanent shackle; its ultimate goal is

self-regulated language use and inquiry.

5.5. Implications for Teachers

To implement the proposed framework, teachers are advised to: (a) start with small-scale, low-stakes PBL units to build confidence; (b) collaborate with colleagues to co-design tasks that tightly align language targets with inquiry goals; (c) use the assessment rubric formatively to provide ongoing feedback; and (d) engage in reflective practice to continuously adjust the dynamic balance between language and inquiry for their specific student groups.

6. Conclusion

The “language-inquiry” binary opposition in primary school English PBL results from metaphysical thinking and deviation from the essence of the discipline. This study has demonstrated that the disciplinary essence of English provides the boundary and direction, while PBL’s connotation provides the pathway and field; their dialectical unity is the prerequisite for realizing PBL’s educational value. The practical framework of cognitive reconstruction, design anchoring, evaluation innovation, and support assurance, together with the concrete case and rubric, translates this dialectical unity into actionable strategies. The primary contribution of this study, compared with prior work that treats language and inquiry in isolation, lies in its explicit theorization of their dialectical relationship and the provision of operational tools for classroom implementation. Future research could conduct longitudinal empirical studies to test the framework’s effectiveness and further refine the assessment instruments. Only when language and inquiry are truly integrated can primary school English PBL fulfill the curriculum’s vision of “integrating learning with thinking and application with creation” [18].

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